User Manual: AI-Powered Financial Health Advisor Capstone Project

HEMACHANDRA HD

1. Setup Instructions

Requirements

Install the required dependencies:

```
pip install -r requirements.txt
```

Required environment variables:

- `GOOGLE_API_KEY` (used to access Gemini 1.5 Pro)

Create a `.env` file:

```
GOOGLE_API_KEY=" GOOGLE_API_KEY"
```

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2. How to Run the Application

From the root directory, run:

```
streamlit run streamlit_app.py
```

Access the app at:
`http://localhost:8501` (local)
`http://<your-ec2-public-ip>:8501` (deployed)

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3. Uploading Financial Documents

Supported formats:

- **PDF**, **CSV**, or **Plain Text**

Document types handled:

- Paystubs
- Bank Statements
- Tax Returns

Once uploaded:

- Document type is auto detected
- Parsed and sent to Gemini via LangChain
- Coaching report and chatbot responses are generated

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4. Application Flow

1. 📂 **Upload** documents in the sidebar
2. 📄 **Preview** raw text and extracted fields
3. 📊 **Review coaching report** generated from Gemini
4. 💬 **Ask questions** via the chatbot powered by LangChain

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5. Deployment Guide (AWS EC2)

1. Launch an EC2 instance (Ubuntu)
2. Open port 8501 in the security group
3. SSH into the instance:

```
ssh -i your_key.pem ubuntu@your-ec2-public-ip
```

4. Clone your project repo and run:

```
streamlit run streamlit_app.py
```

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6. Gemini + LangChain Integration

- Uses Gemini 1.5 Pro for document understanding and financial coaching
- LangChain is used to:
  - Structure prompt chaining
  - Enable conversational chatbot
  - Parse structured JSON outputs

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7. Sample Inputs and Expected Output

- Uploaded paystub → Extracts: Gross Pay, Net Pay, Taxes
- Uploaded bank statement → Summarizes: Expenses, Recurring Charges
- Generates coaching advice:
  - Savings rate analysis
  - Emergency fund recommendations
  - Debt reduction tips

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8. Troubleshooting

- **Gemini not responding?** → Check GOOGLE_API_KEY and internet access.
- **App not launching?** → Check that Streamlit is installed, and you are using the correct virtual environment.
- **Wrong document type detected?** → Try manual override by renaming or uploading one file at a time.